

Cover Letter — Quantum Information Processing

14 July 2026

Dear Editors of *Quantum Information Processing*,

On behalf of my co-authors, I am pleased to submit our original research article, “**Quantum Oracle Sketching Robustness under Structured Non-IID Data Streams**”, for consideration in *Quantum Information Processing*.

What the paper does. Quantum oracle sketching (Zhao et al., 2026) provides proven exponential memory separations for streaming classification and dimension reduction, with a correlated-data extension in which sample complexity is multiplied by the repetition number of the data-generation process. Our paper contributes the first systematic robustness mapping of this framework under structured, practically motivated correlation families. We define operational proxies for the framework’s correlation parameters (refreshing time τ , repetition number r), evaluate five non-IID regimes against three classical memory-bounded baselines (10 seeds, 95% CIs, paired Wilcoxon tests), quantify where analytic proxies are conservative relative to empirical estimates (up to $125\times$ for long-memory streams), and locate the proxy-model advantage crossover at Markov strength $\rho \approx 0.88$. The study is transparently proxy-based: it neither re-proves nor simulates the underlying quantum algorithm, and every divergence between our heuristic model and the exact framework is stated explicitly next to the corresponding equations.

Fit to Quantum Information Processing. The paper addresses the practical applicability of a prominent quantum-advantage result — the robustness of quantum streaming advantage beyond IID assumptions — a question squarely within QIP’s scope of quantum algorithms and their behaviour under realistic conditions, and it connects to the quantum streaming separations line (Kallaugh et al., 2021, 2025).

Declarations. The manuscript is original, unpublished, and not under consideration elsewhere. All authors approved the submission. No competing interests; no external funding. A generative-AI-assistance declaration is included per Springer policy. All code and data are open (Zenodo DOI: 10.5281/zenodo.19831893, MIT licence). The manuscript is submitted in a standard LaTeX layout under Springer’s “Your Paper Your Way” policy; we will convert to the Springer class at revision.

Suggested reviewers: [to be entered by the authors in the submission system].

Thank you for considering our work.

Sincerely, on behalf of all authors,

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